Year 10 Foundation

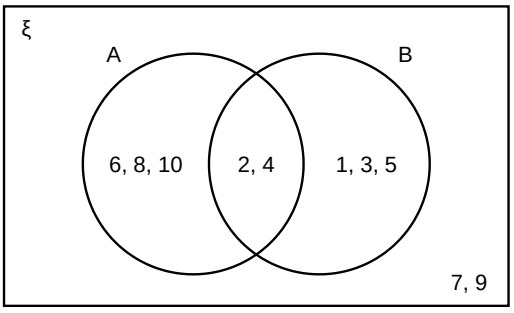
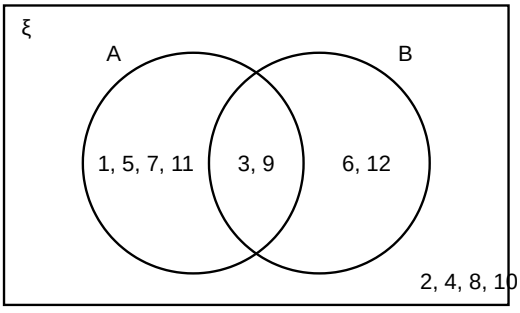
REVISION PACK

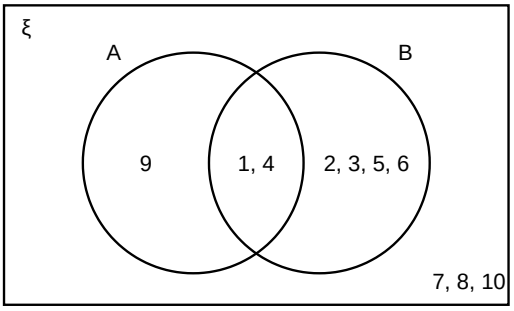
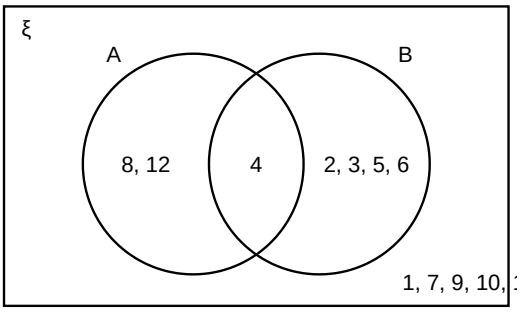
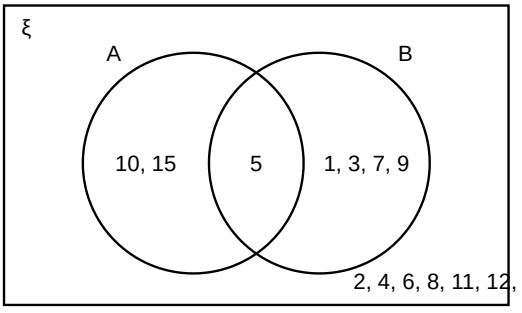
Sets 1

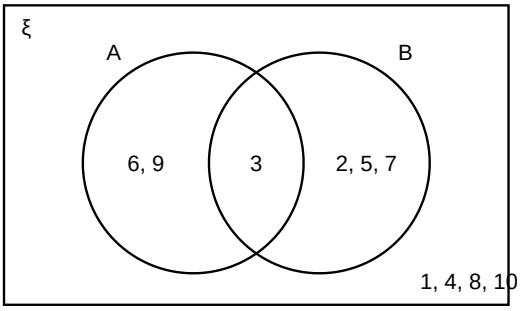
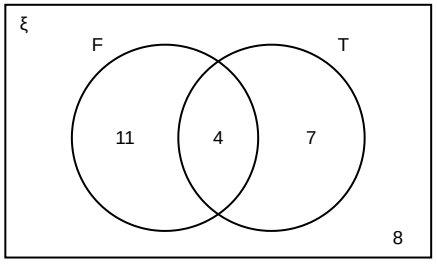
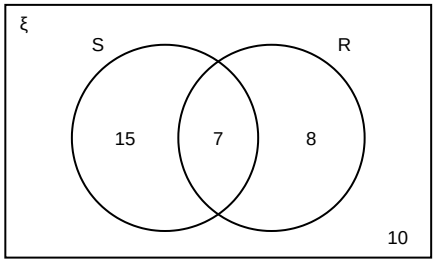
ANSWERS

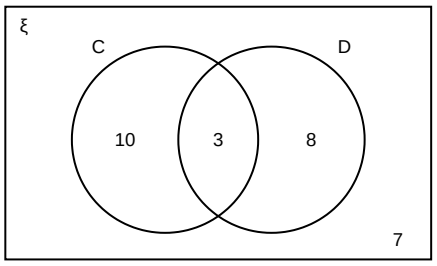
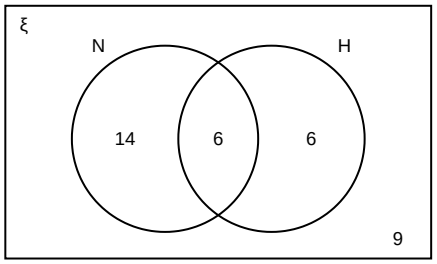
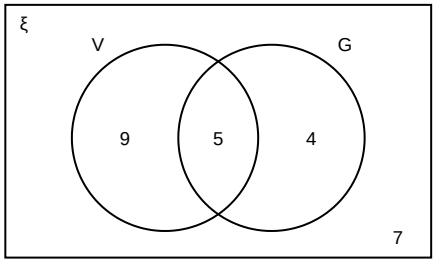
Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

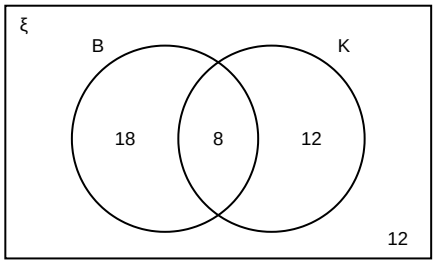
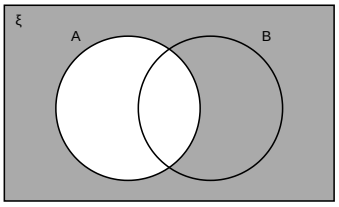
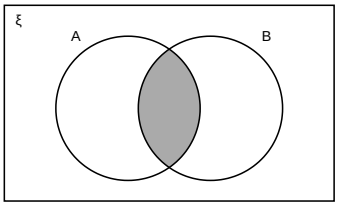
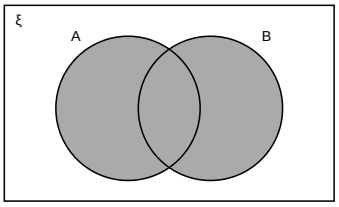
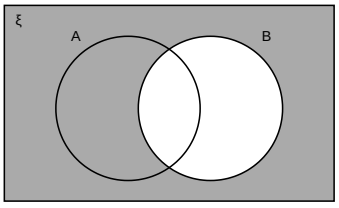
Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2)

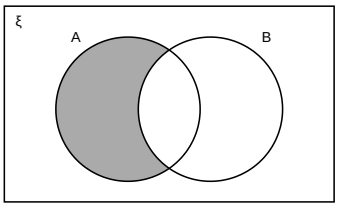
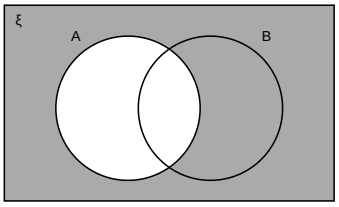
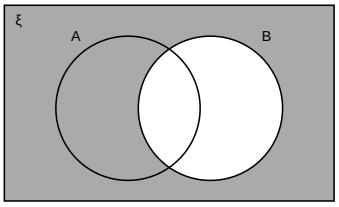
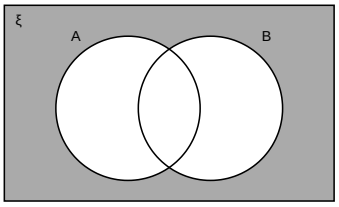
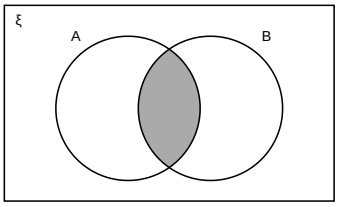
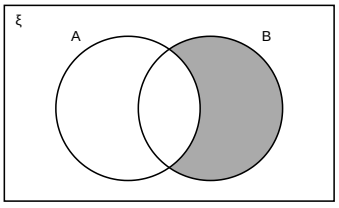
Q	Working	Answer
completing a Venn diagram		
1.1	**(a)** **(b)** **(c)** **(d)** $A' = \{1, 3, 5, 7, 9\}$ 	**(a) see the diagram** **(b) {2, 4}** **(c) {1, 2, 3, 4, 5, 6, 8, 10}** **(d) 5**
1.2	**(a)** **(b)** **(c)** **(d)** $A' = \{2, 4, 6, 8, 10, 12\}$ 	**(a) see the diagram** **(b) {3, 9}** **(c) {1, 3, 5, 6, 7, 9, 11, 12}** **(d) 6**

1.3	**(a)** **(b)** **(c)** **(d)** $A' = \{2, 3, 5, 6, 7, 8, 10\}$  	**(a) see the diagram** **(b) {1, 4}** **(c) {1, 2, 3, 4, 5, 6, 9}** **(d) 7**
1.4	**(a)** **(b)** **(c)** **(d)** $A' = \{1, 2, 3, 5, 6, 7, 9, 10, 11\}$  	**(a) see the diagram** **(b) {4}** **(c) {2, 3, 4, 5, 6, 8, 12}** **(d) 9**
1.5	**(a)** **(b)** **(c)** **(d)** $A' = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14\}$  	**(a) see the diagram** **(b) {5}** **(c) {1, 3, 5, 7, 9, 10, 15}** **(d) 12**

1.6	**(a)** **(b)** **(c)** **(d)** $A' = \{1, 2, 4, 5, 7, 8, 10\}$  	**(a) see the diagram** **(b) {3}** **(c) {2, 3, 5, 6, 7, 9}** **(d) 7**
problems with two sets		
1.7	**(a)** $15 - 4 = 11$ $11 - 4 = 7$ $30 - (11 + 4 + 7) = 8$ **(b)** $11 + 7 = 18$  	**(a) see the diagram** **(b) 18**
1.8	**(a)** $22 - 7 = 15$ $15 - 7 = 8$ $40 - (15 + 7 + 8) = 10$ **(b)** $15 + 8 = 23$  	**(a) see the diagram** **(b) 23**

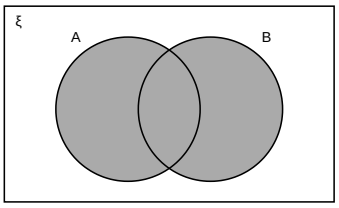
1.9	**(a)** $13-3=10$ $11-3=8$ $28-(10+3+8)=7$ **(b)** $10+8=18$ 	**(a) see the diagram** **(b) 18**
1.10	**(a)** $20-6=14$ $12-6=6$ $35-(14+6+6)=9$ **(b)** $14+6=20$ 	**(a) see the diagram** **(b) 20**
1.11	**(a)** $14-5=9$ $9-5=4$ $25-(9+5+4)=7$ **(b)** $9+4=13$ 	**(a) see the diagram** **(b) 13**

1.12	**(a)** $26 - 8 = 18$ $20 - 8 = 12$ $50 - (18 + 8 + 12) = 12$ **(b)** $18 + 12 = 30$ 	**(a) see the diagram** **(b) 30**
shading regions		
1.13	(a) A'  (b) $A \cap B$ 	
1.14	(a) $A \cup B$  (b) B' 	

1.15	(a) $A \cap B'$  (b) A'  	
1.16	(a) B'  (b) $(A \cup B)'$  	
1.17	(a) $A \cap B$  (b) $A' \cap B$  	

1.18

(a) $A \cup B$



(b) $A \cap B'$

